

AI-Based Radius and Ulna Bone Segmentation for Wrist Fracture Surgical Planning

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Abstract

This report presents the development and validation of efficient custom deep learning architectures for automated radius and ulna bone segmentation from CT scans for pre-operative surgical planning of distal radius fractures. The project addresses computational efficiency limitations inherent to nnU-Net, which Specifix currently employs for bone segmentation. While nnU-Net represents the state-of-the-art in medical image segmentation, its substantial computational requirements present challenges for clinical deployment and rapid development iteration. To establish a performance benchmark, our team implemented nnU-Net following standard protocols, achieving 90 % test Dice (post-processed) with 31M parameters, 63h training time, and batch size of 2. Raw nnU-Net predictions achieved 94.1 % with 216 connected components; post-processing reduced this to 2.4 components while maintaining clinical accuracy. Our custom ResU-Net architectures achieve clinical-grade segmentation accuracy while dramatically reducing these computational requirements. The best-performing model delivers test Dice Similarity Coefficient of 86.9 % using only 3.2M parameters (90 % reduction), with training converging in 7.0 h (89 % reduction) and inference completing in under 4 min. This performance approaches our nnU-Net baseline within 3.1 percentagepoints while delivering practical deployment advantages through reduced hardware requirements, faster development iteration, and lower operational costs. The system maintains processing time well under 10 min per case, representing an 85 % to 90 % improvement over 30 min to 45 min manual segmentation workflows. Beyond technical performance, the project establishes comprehensive regulatory and ethical frameworks positioning Specifix for commercial deployment as a Class IIa medical device under EU regulations. The research validates that purpose-built architectures optimized for specific anatomical structures can match state-of-the-art generic framework performance with substantially improved computational efficiency, supporting more accessible clinical deployment.

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1. Executive Summary

This report presents the development and validation of efficient custom deep learning architectures for automated radius and ulna bone segmentation from computed tomography (CT) scans, designed to support pre-operative surgical planning for distal radius fractures. The 18-week academic research project was conducted in collaboration between Breda University of Applied Sciences and Specifix B.V., a medical technology company focused on AI-driven orthopedic surgical planning solutions.

Distal radius fractures represent one of the most common orthopedic injuries worldwide, accounting for approximately 8–18% of all fractures globally. The current manual segmentation process requires over 30 minutes per case, creating significant bottlenecks in surgical planning workflows. Our solution addresses this critical gap by developing custom neural network architectures that achieve clinical-grade accuracy while dramatically reducing computational requirements compared to existing state-of-the-art frameworks.

The project delivered a validated research prototype demonstrating that purpose-built architectures can approach the performance of computationally intensive frameworks like nnU-Net while requiring only a fraction of the resources. The best custom model achieved 86.9 % test Dice score using just 3.2M parameters—a 90% reduction compared to nnU-Net’s 31M parameters—with training completing in 7.0 hours versus 63 hours. These findings validate the commercial viability of efficient automated segmentation for clinical deployment.

2. Introduction

2.1. Clinical Background and Problem Context

Distal radius fractures represent one of the most common orthopedic injuries worldwide, accounting for approximately 8–18% of all fractures globally (1). The clinical burden extends across multiple patient demographics, with fractures comprising up to 25% of pediatric injuries and 18% of fractures in elderly populations (2). The global epidemiological impact translates to age-standardized incidence rates of 179 cases per 100,000 individuals worldwide (3), creating substantial demands on healthcare systems.

Current pre-operative surgical planning for these fractures requires extensive manual analysis of CT scans. Surgeons must spend 30–45 minutes per case performing critical tasks including fracture identification and classification, anatomical structure delineation, surgical approach planning, and three-dimensional model generation (4).

This manual workflow creates several cascading problems throughout the healthcare delivery chain.

2.2. Current Workflow Inefficiencies

The existing manual segmentation process creates significant operational bottlenecks that impact both clinical efficiency and patient outcomes. Surgeons spend over 30 minutes per case on routine segmentation tasks, diverting surgical expertise away from direct patient care and complex clinical decision-making. This time-intensive requirement creates treatment delays, resource misallocation, and increased healthcare costs.

The manual approach introduces substantial inter-operator variability affecting surgical planning consistency. Segmentation quality depends heavily on individual operator experience, attention to detail, and subjective interpretation of anatomical boundaries. This variability can lead to inconsistent surgical planning and potentially suboptimal patient outcomes. Automation offers the potential to eliminate this variability while dramatically reducing processing time.

2.3. Market Landscape and Strategic Opportunity

The current orthopedic surgical planning market demonstrates a clear gap in automated segmentation capabilities. Brainlab (5) specializes in surgical navigation and operating room integration but does not provide automated CT-based segmentation. PeekMed (6) focuses on pre-operative planning and implant templating primarily using X-ray images. Materialise (7) leads in three-dimensional printing for patient-specific implants but depends on accurate digital anatomy provided upstream.

The only FDA-approved AI tool in this domain, OsteoDetect (8), analyzes two-dimensional wrist radiographs but cannot handle three-dimensional CT-based volumetric segmentation. This regulatory approval demonstrates that AI-powered fracture analysis tools can meet clinical and regulatory standards while highlighting the specific gap in volumetric segmentation capabilities that Specifix addresses.

2.4. Current State and Benchmark Establishment

Specifix currently employs nnU-Net (9) for radius bone segmentation in their surgical planning workflow. While nnU-Net represents the state-of-the-art in medical image segmentation, its computational requirements present practical challenges for clinical deployment and rapid development iteration. To understand the framework’s behavior and establish a performance benchmark for evaluating custom architectures, our team trained nnU-Net following standard protocols on the project dataset.

2.5. Project Objectives and Scope

This 18-week academic research project was conducted in collaboration between Breda University of Applied Sciences and Specifix B.V. The primary objective was to develop and validate custom automated segmentation architectures that could match or approach nnU-Net’s clinical accuracy while dramatically reducing computational requirements in three key areas: computational expense (parameter count and memory usage), training time (enabling faster development iteration), and inference time (supporting efficient clinical workflows).

To establish a rigorous performance benchmark, the team first trained nnU-Net following standard protocols on the project dataset. This benchmark served as the performance ceiling for evaluating custom architectures, with the goal of achieving Dice Similarity Coefficient (DSC) above 85 % and complete case processing under 10 min while substantially reducing nnU-Net’s computational demands. Success would demonstrate that specialized models can provide competitive clinical accuracy with dramatically improved efficiency, potentially offering Specifix a more computationally accessible alternative to their current approach. For comprehensive project planning details, refer to [Appendix D](#) (Business Requirements Document).

3. Problem Definition

3.1. Clinical Workflow Problem

The fundamental problem addressed by this project is the time-intensive nature of manual radius bone segmentation in surgical planning workflows. Surgeons currently spend 30–45 minutes per case manually delineating radius anatomy from CT scans. This manual process involves identifying all bone fragments (particularly in complex fractures), distinguishing radius from adjacent ulna bone, tracing anatomical boundaries across hundreds of CT slices, and generating three-dimensional models suitable for surgical planning.

This workflow creates multiple clinical challenges. Treatment delays occur as segmentation bottlenecks extend time between imaging and surgical intervention. Resource misallocation diverts highly trained surgical staff from direct patient care to routine image processing. Inter-operator variability introduces inconsistencies in anatomical boundary determination affecting surgical planning quality. These factors combine to increase healthcare costs while potentially compromising patient outcomes.

3.2. Technical Segmentation Challenges

CT scans of fractured wrists present several technical complexities that automated segmentation systems must address. Fractured anatomy disrupts normal spatial relationships that algorithms typically rely upon—the radius bone may be broken into multiple fragments with varying displacement patterns requiring the system to recognize and merge discontinuous structures. Adjacent anatomical structures, particularly the ulna, share similar density characteristics (both are cortical bone visible as bright regions in CT) and must be accurately distinguished despite minimal intensity differences.

Simple fractures with minimal displacement present relatively straightforward segmentation scenarios. Complex multi-fragment fractures with significant displacement challenge algorithms to maintain anatomical coherence across disconnected regions. Severe comminution (bone shattered into numerous small fragments) may exceed current model capabilities, requiring human intervention.

3.3. Dataset and Computational Constraints

The available dataset comprises 162 CT scan cases provided by Specifix through their clinical partnership with Radboud University Medical Center. This relatively small dataset size presents challenges for deep learning approaches that typically benefit from thousands of training examples. The data includes both fractured wrist cases (F-cases and P-cases) and contralateral healthy scans (CL-cases), providing comprehensive anatomical coverage but limited sample size for robust model generalization.

Cases were stored in NRRD format with fragmented files representing individual bone structures (M0, M1, etc. for different fragments), requiring sophisticated preprocessing to merge fragments and extract clean training data. Data quality assessment identified several structural inconsistencies requiring correction before model training could commence.

The computational efficiency challenge centers on nnU-Net's substantial resource requirements. While nnU-Net achieves excellent segmentation accuracy, its computational demands create practical deployment barriers including high cloud computing costs during development, slow iteration cycles limiting rapid experimentation, hardware requirements that may limit deployment sites, and processing speeds that could create workflow bottlenecks in high-volume clinical settings. The challenge was to develop purpose-built architectures that maintain clinical-grade accuracy while operating within substantially reduced resource constraints.

For detailed problem definition and project scope, refer to [Appendix D](#) (Business Requirements Document).

4. Data Preparation

4.1. Overview of Preprocessing Pipeline

The raw dataset required substantial preprocessing to transform fragmented NRRD files into training-ready format. The team developed an 8-step preprocessing pipeline addressing data quality issues, standardizing spatial and intensity properties, and creating appropriately sized training samples. The pipeline supports multiple bone selection modes (radius only, ulna only, or multiclass) to accommodate different research objectives.

4.2. Pipeline Steps

Step 1: Data Cleaning. Contralateral (CL) folders were extracted as independent cases, creating new case identifiers (e.g., F037/CL became F037CL). This restructuring made healthy bone references directly accessible as training samples, increasing the dataset from 114 to 162 effective cases. Files in nested subdirectories were moved to appropriate parent folders, invalid cases with missing required files were removed, and unnecessary folders were cleaned.

Step 2: Label Merging and Naming Standardization. Multiple mask fragments per case (M0, M1, and additional labeled fragments representing different bone pieces) were merged into single consolidated masks. All files were renamed following consistent conventions: CT scans as {case}_scan.nrrd and masks as {case}_{mode}.nrrd.

Step 3: Voxel Spacing Standardization. All volumes were resampled to uniform 0.6 mm isotropic spacing. CT scans used B-Spline interpolation to preserve image detail while masks used Nearest Neighbor interpolation to preserve integer label values.

Step 4: Intensity Normalization. Intensity statistics (percentiles, mean, standard deviation) were calculated from bone regions only across the entire dataset. Intensities were clipped to 1st–99th percentiles to remove outliers, then z-score normalized to approximately mean=0, std=1 distribution.

Step 5: Train/Validation/Test Split. The dataset was divided using 70/15/15 split ratios with stratification ensuring each split contained proportional representation of case categories (F-cases, P-cases, CL-cases). Complete unpatched volumes for all test cases were preserved for full-volume evaluation.

Step 6: Patch Extraction. Full 3D volumes were extracted into $128 \times 128 \times 128$ voxel patches using sliding window approach with configurable overlap, maintaining paired scan and mask data.

Step 7: Intelligent Patch Filtering. To address severe class imbalance (bone structures occupy small fraction of total volume), patches were categorized as foreground (contain labeled bone), unlabeled bone (high intensity but no labels), or empty (no bone content). All foreground patches were retained while unlabeled and empty patches were downsampled. The test set was NOT filtered to ensure unbiased evaluation.

Step 8: HDF5 Batching. Filtered patches were packaged into HDF5 batch files for efficient data loading during training, with 128 patches per file and no compression to optimize read speed.

For comprehensive preprocessing methodology and technical details, refer to [Appendix E](#) (Data Preparation Report).

5. Proposed Solution

5.1. Overall Technical Approach

Our solution employs purpose-built deep learning architectures optimized specifically for fractured wrist bone segmentation, aiming to match the clinical accuracy of nnU-Net while dramatically reducing computational requirements. The approach positions our nnU-Net baseline implementation as the performance benchmark to beat in terms of computational expense, training time, and inference speed, while maintaining competitive accuracy.

The methodology systematically explored multiple neural network architectures to identify optimal configurations balancing clinical ac-

curacy against computational efficiency. The team trained nnU-Net following standard protocols to establish the performance ceiling and developed custom architectures designed to achieve comparable accuracy with substantially reduced resource requirements. Key strategies included selective data augmentation (evaluated empirically), architectural innovations including residual connections optimized for moderate parameter counts, and comprehensive evaluation frameworks assessing both clinical accuracy and computational resource requirements against the nnU-Net baseline.

5.2. Architectural Variants Evaluated

The project systematically evaluated multiple neural network architectures. The baseline three-dimensional U-Net (10) represented the established standard for volumetric medical image segmentation. ResU-Net architectures incorporated residual connections enabling deeper networks that learn residual mappings rather than direct transformations. MultiScale ResU-Net explored whether additional context improves segmentation accuracy by processing input at multiple resolutions simultaneously.

The nnU-Net framework (9) provided a critical performance benchmark, with this self-configuring approach automatically determining optimal preprocessing, architecture configuration, and training strategies based on dataset characteristics. V-Net (11) served as an evaluated architecture with residual connections and Dice loss optimization demonstrating particular effectiveness for imbalanced medical segmentation tasks, though it was discontinued after Phase 1 due to unfavorable cost-benefit ratio compared to ResU-Net.

5.3. Data Augmentation Strategy

Given the limited dataset size of 162 cases, data augmentation strategies were explored across multiple model configurations. Augmentation techniques evaluated included spatial augmentations (random rotations, scaling transformations, elastic deformations, mirroring operations) and intensity augmentations (brightness/contrast adjustments, Gaussian noise, gamma transformations).

Critically, the best-performing models for both radius-only and multiclass segmentation tasks were trained *without* augmentation. The radius-only best model and multiclass best model both achieved superior performance without augmentation, suggesting that for this specific dataset and task, the preprocessing pipeline (voxel spacing standardization, intensity normalization) provided sufficient data consistency, and augmentation may have introduced unnecessary variability that hindered optimal convergence.

5.4. Training Procedures and Optimization

Model training followed rigorous protocols designed to maximize performance while enabling fair comparison across architectural variants. The optimization strategy employed Adam and AdamW optimizers with learning rate scheduling, mixed precision training enabling larger batch sizes (when applicable), and gradient clipping preventing instability. Loss functions combined Dice loss directly optimizing the evaluation metric with cross-entropy and focal loss providing complementary gradients, while deep supervision from intermediate layers improved gradient flow in deep architectures.

For detailed problem definition, proposed solution, and acceptance criteria, refer to [Appendix D](#) (Business Requirements Document).

6. Evaluation Criteria

6.1. Primary Metric: Dice Similarity Coefficient

The Dice Similarity Coefficient (DSC) served as the primary evaluation metric, quantifying overlap between predicted and ground truth segmentation masks. Values range from 0 (no overlap) to 1 (perfect agreement), with coefficients above 85 % representing clinical acceptability based on medical imaging literature and project requirements. DSC directly measures the core objective: accurate identification of bone tissue.

6.2. Computational Efficiency Metrics

Computational efficiency metrics documented resource requirements essential for practical clinical deployment. Parameter counts quantified model complexity and memory requirements, directly impacting deployment hardware specifications. Training duration measured end-to-end time from initialization to convergence, impacting iterative development cycles and model improvement velocity. Inference speed determined processing time per case, with the target of under 10 min directly relevant for clinical workflow integration.

6.3. Clinical Workflow Metrics

Processing time comparison against manual segmentation (30 min to 45 min) established the efficiency value proposition. Consistency assessment evaluated inter-operator variability elimination through automated processing. Reproducibility validation ensured identical results across repeated processing of the same case. Three-dimensional model quality assessed suitability for dimensional measurements and surgical planning through visual inspection.

For complete evaluation metrics and their clinical significance, refer to [Appendix B](#) (Performance Metrics Definitions).

7. Research Methodology

7.1. Dataset Acquisition and Characteristics

The project utilized 162 CT scans provided by Specifix through their clinical partnership with Radboud University Medical Center. The dataset included both fractured wrist cases and contralateral healthy scans, providing comprehensive coverage of anatomical variation. Ground truth annotations were provided by clinical experts, establishing validated reference standards for model training and evaluation.

7.2. Experimental Design

The research followed a systematic experimental methodology conducting over 150 training runs across multiple architectural variants, hyperparameter configurations, and data processing strategies. Early experiments established baseline performance and validated preprocessing approaches. Mid-project iterations focused on architectural refinements and augmentation strategies. Final experiments optimized the most promising architectures and validated performance on held-out test data.

Infrastructure challenges including GPU memory constraints and Docker shared memory limitations required careful resource management and influenced architectural design choices. The team implemented standardized evaluation protocols ensuring fair comparison across all architectural variants through consistent preprocessing, identical train/validation/test splits, standardized hyperparameter ranges, and comprehensive logging of all experiments.

7.3. Evaluation Framework

Case-level evaluation required careful attention to inference procedures. The team implemented sliding window inference with Gaussian weighting to handle full patient volumes that exceeded GPU memory capacity. Post-processing steps including connected component analysis and morphological operations improved clinical usability of predicted masks. Visual inspection using 3D Slicer (12) software enabled qualitative assessment of segmentation quality and failure mode identification.

The comparative analysis framework systematically compared different approaches through baseline comparisons establishing performance benchmarks, architecture comparisons evaluating impact of specific design choices, nnU-Net comparison providing critical context as state-of-the-art benchmark, and augmentation studies isolating the impact of data augmentation strategies.

8. Results and Analysis

8.1. nnU-Net Benchmark Results

The team implemented and trained nnU-Net following standard protocols to establish a rigorous performance benchmark for evaluating custom architectures and to understand the computational characteristics of the framework that Specifix employs. The self-configuring framework automatically analyzed dataset characteristics and determined optimal architecture settings, preprocessing parameters, and training strategies without manual tuning.

Our benchmark implementation achieved validation Dice of 94.1 % and test Dice of 90.0 % (post-processed) for radius segmentation, demonstrating state-of-the-art performance that validates both dataset quality and task feasibility. Raw predictions achieved 94.1 % test Dice but contained 216 connected components indicating severe fragmentation. Post-processing steps including largest component selection, hole filling, and morphological closing reduced this to an average of 2.4 components per case, yielding clinically usable segmentations while maintaining 90 % test performance.

The architecture utilized 31M parameters and required 63 h of training time (1000 epochs) on available GPU hardware with batch size of 2. Inference processing took approximately 40 s per case. This comprehensive benchmark established clear targets: custom architectures needed to achieve test Dice above 85 % (clinical threshold) while substantially reducing the 31M parameters, 63 h training time, and batch size constraints that present practical deployment challenges.

8.2. Custom Architecture Performance

Table 1 summarizes the performance comparison between the nnU-Net benchmark and the developed custom architectures.

Table 1. Top Performing Model Architectures

Model	Params (M)	Test Dice (%)	Runtime (hrs)
nnU-Net 3D	31.0	90.0	63.0
ResU-Net v4 (24f)	3.2	86.9	7.0
ResU-Net v5 (20f)	2.2	86.2	8.6
V-Net-Res	11.6	82.4	12.5
3D U-Net (32f)	12.9	77.6	2.7

ResU-Net Version 4 (24 initial features) provided excellent balance of accuracy and efficiency among custom architectures, successfully approaching our nnU-Net baseline’s clinical accuracy while dramatically reducing computational requirements. The model achieved validation Dice of 93.2 % and test Dice of 86.9 % using only 3.2M parameters, a 90% parameter reduction compared to our nnU-Net implementation. Training converged in 7.0 h (75 epochs), representing an 89% reduction in training time. Per-case analysis showed 18 of 23 test cases (78.3%) met the clinical threshold of $DSC \geq 0.85$, with 14 cases (60.9%) achieving excellent performance ($DSC \geq 0.90$).

ResU-Net Version 5 (20 initial features) further optimized computational efficiency. With only 2.2M parameters, this model achieved test Dice of 86.2 %, representing a 93% parameter reduction compared to our nnU-Net baseline. Training time of 8.6 h represented an 86% reduction compared to nnU-Net’s 63 h requirement.

The critical finding was that these custom architectures achieved clinical-grade accuracy (exceeding the 85 % threshold) with only 3.1-3.8 percentage points performance sacrifice compared to our nnU-Net baseline, while delivering dramatic improvements across all computational metrics.

8.3. Computational Efficiency Analysis

A critical project objective was demonstrating that clinical-grade accuracy could be achieved with substantially reduced computational requirements. Table 2 summarizes the efficiency comparisons.

Table 2. Computational Efficiency: Custom Architecture vs. nnU-Net Baseline

Model	Params (M)	Train (hrs)	Infer (min)	Dice (%)
nnU-Net	31.0	63.0	6.5	90.0
ResU-Net v4	3.2	7.0	3.5	86.9
ResU-Net v5	2.2	8.6	3.5	86.2
Reduction (v4)	90%	89%	46%	−3.1%
Reduction (v5)	93%	86%	46%	−3.8%

The best custom architectures utilized 2.2-3.2M parameters compared to our nnU-Net baseline with 31M parameters, representing 90-93% parameter reductions. Training duration showed 86-89% time reductions from 63 h to 7 h to 8.6 h, dramatically improving development iteration speed. Inference time improved from nnU-Net’s 40 s to 3 min to 4 min for complete pipeline processing, well under the 10 min target.

The critical finding was that these dramatic efficiency gains required only 3.1-3.8 percentage points accuracy sacrifice, falling within acceptable tolerance for many clinical applications. This validates that purpose-built architectures deliver substantial practical advantages over generic frameworks.

8.4. Multiclass Segmentation Results

The project extended from binary radius-only segmentation to multiclass segmentation of both radius and ulna bones. The best multiclass model (ResU-Net with 8 initial features) achieved overall test Dice of 80.7 %, with class-specific performance showing radius at 75.7 % and ulna at 85.7 %. The model utilized 2.9M parameters and trained in 3.0 h (50 epochs).

The multiclass task demonstrated increased complexity, with per-class analysis revealing that ulna consistently outperformed radius by approximately 10 percentage points. This performance difference reflects anatomical characteristics: ulna is typically larger with more consistent morphology, while radius exhibits greater variability and is more frequently fractured. Of 19 test cases, 13 (68.4%) achieved acceptable quality ($DSC > 0.75$), demonstrating practical value for dual-bone surgical planning workflows.

8.5. Critical Technical Findings

Several critical insights emerged from the extensive experimentation. The impact of data augmentation was unexpected: the best-performing models for both segmentation tasks were trained *without* augmentation, suggesting that for this specific dataset, the preprocessing pipeline provided sufficient data consistency, and augmentation may have introduced distribution shifts between training and test data.

Model capacity required careful optimization. Models in the 2-4 million parameter range achieved near-optimal accuracy while maintaining practical efficiency, validating that purpose-built architectures optimized for specific tasks can match generic framework performance with substantially reduced complexity.

Residual connections enabled deeper networks without degradation, improving gradient flow during training. The ResU-Net architecture with 3.2M parameters consistently outperformed larger architectures including standard U-Net (12.9M) and V-Net (11.6M), demonstrating superior parameter efficiency through improved gradient propagation.

8.6. Performance Across Clinical Scenarios

Models performed consistently across both fractured and healthy contralateral anatomy. Simple fractures with minimal displacement showed the highest segmentation accuracy while complex multi-fragment fractures with significant displacement presented greater challenges. One extreme case with severe fragmentation achieved

DSC below 0.50, demonstrating model limitations on extreme fracture patterns requiring manual intervention.

8.7. Comparison with Manual Segmentation

Manual segmentation requires 30–45 minutes per case while automated processing completes in under 10 minutes, representing an 85–90% time reduction. Automated systems produce consistent results eliminating operator-dependent quality variation and provide perfect reproducibility. The combination validates the business case for automated segmentation.

For comprehensive model development details including architectural variants, training configurations, hyperparameter exploration, and per-case performance analysis, refer to [Appendix G](#) (Model Development Report). For complete training logs documenting all 150+ experimental runs, refer to [Appendix F](#) (Model Training Logs). For baseline architecture comparison and selection rationale, refer to [Appendix H](#) (Baseline Comparison Report).

9. Regulatory and Ethical Framework

9.1. Regulatory Classification

The system is classified as a high-risk AI system under EU AI Act Article 6.1 and Class IIa medical device under Medical Device Regulation (MDR) Rule 11, Annex VIII. High-risk classification triggers mandatory compliance with risk management systems, data governance and bias assessment, technical documentation, transparency requirements, human oversight mechanisms, and accuracy validation. Class IIa classification requires conformity assessment by notified body, CE marking, clinical evaluation, post-market surveillance, and quality management system per ISO 13485 (13).

9.2. Ethical Framework Application

The project applied four complementary international ethical frameworks including UNESCO Recommendation on Ethics of AI providing ten core principles, EU High-Level Expert Group Guidelines establishing trustworthy AI requirements, WHO Standards for Medical AI emphasizing safety and effectiveness, and WMA Physician Professional Duties requiring independent professional judgment.

9.3. Risk Assessment and Mitigation

Comprehensive risk assessment identified segmentation accuracy risks, automation bias risks, privacy and security risks, fairness and bias risks from known dataset demographic biases, and edge case failure risks. Each risk category was addressed through specific mitigation strategies including validation protocols, human-in-the-loop design, encrypted storage, surgeon oversight, and comprehensive documentation.

9.4. Human-AI Collaboration Model

The human-AI collaboration model balances automation efficiency with clinical accountability through mandatory surgeon review of all outputs, manual correction capabilities, and explicit approval workflows ensuring clinical expertise validates AI-generated anatomical boundaries. This design simultaneously addresses regulatory requirements for human oversight, ethical principles emphasizing human agency, and professional standards requiring personal verification of clinical information.

For complete regulatory and ethical analysis including detailed compliance requirements, risk assessment matrices, and mitigation strategies, refer to [Appendix I](#) (Regulatory and Ethical Compliance).

10. Conclusions

10.1. Primary Objectives Achievement

The project successfully achieved all primary objectives, demonstrating that custom architectures can approach nnU-Net performance

while dramatically reducing computational requirements. The best custom architectures (ResU-Net v4 and v5) achieved test DSC of 86.2 %-86.9 %, exceeding the 85 % clinical acceptability threshold and approaching within 3.1-3.8 percentage points of our nnU-Net baseline's 90 % benchmark.

Complete case processing consistently completed in under 10 min, exceeding the efficiency target and demonstrating 85–90% time reduction compared to manual workflows. More critically, the custom architectures beat our nnU-Net baseline across all computational metrics: 90-93% parameter reduction, 86-89% training time reduction, and substantially faster inference enabling higher throughput clinical workflows.

10.2. Technical Contributions

The research makes several broader contributions to medical imaging AI. Systematic evaluation of data augmentation strategies revealed that clinical-grade performance is achievable without augmentation if appropriate preprocessing is employed, challenging conventional assumptions about augmentation necessity for small datasets. The finding that moderate-capacity specialized architectures in the 2-4 million parameter range outperform both very small and very large models challenges conventional assumptions about model scaling for small medical datasets.

The demonstration that residual connections provide superior parameter efficiency compared to simply increasing model size provides practical insights for similar medical imaging projects facing data constraints.

10.3. Project Value and Strategic Impact

This project validates Specifix's strategy of addressing an upstream gap in the surgical planning workflow that existing competitors do not directly target. Specifix's automated segmentation provides the foundational capability that all downstream planning and manufacturing depend upon, creating opportunities for both standalone value delivery and partnerships within the broader surgical planning ecosystem.

The project delivers substantial value to Specifix through technically validated proof-of-concept ready for commercial development, comprehensive regulatory and ethical framework analysis positioning the system for CE marking, and foundational AI capabilities demonstrating 85-90% time reduction over manual workflows while maintaining clinical-grade accuracy.

The research provides immediate deployment options through validated ResU-Net architectures offering optimal cost-performance balance, established nnU-Net benchmark for applications requiring maximum accuracy, and flexible architecture portfolio enabling deployment strategy customization based on specific clinical requirements and resource constraints.

11. Recommendations and Next Steps

11.1. Model Deployment Recommendations

Based on comprehensive performance evaluation across 150+ training runs and two segmentation tasks, we provide the following deployment recommendations:

Primary Recommendation: Deploy ResU-Net v4 for Production. The ResU-Net Version 4 (24 initial features, 3.2M parameters) achieves the optimal balance between clinical accuracy (86.9 % test Dice) and computational efficiency (7.0 hour training, 3.5 minute inference). This model meets the 85 % clinical acceptability threshold while delivering 90% parameter reduction and 89% training time reduction compared to nnU-Net. Deploy this model for cost-sensitive applications prioritizing deployment flexibility, rapid development iteration, and lower operational costs.

Alternative Recommendation: Deploy nnU-Net for Maximum Accuracy. For applications where absolute accuracy justifies

computational expense, the nnU-Net benchmark (90 % test Dice) remains the superior choice. The 3.1 percentage point accuracy improvement over ResU-Net v4 may be clinically significant for complex fracture cases requiring maximum precision. Deploy nnU-Net when computational resources are available and accuracy is paramount.

Hybrid Deployment Strategy. Consider deploying both models with case-specific selection: use ResU-Net v4 for routine segmentation (fast, efficient, clinically acceptable), and reserve nnU-Net for complex cases requiring maximum accuracy or cases where ResU-Net confidence scores indicate potential challenges.

11.2. Dataset Expansion

Critical gaps requiring attention for commercial deployment include dataset expansion through multi-center collaboration to 1,000 or more cases with comprehensive demographic documentation enabling fairness assessment and subgroup performance validation. Performance validation across patient subgroups with stratified analysis by demographic categories and fracture complexity levels provides necessary fairness evidence for regulatory submission.

11.3. Regulatory Pathway

Formal risk management file consolidation into ISO 14971-compliant format is essential for regulatory submission. Clinical evaluation reports with multi-center prospective studies demonstrating real-world effectiveness represent the critical path for CE marking authorization. ISO 13485-compliant quality management system infrastructure including document control, design controls, and internal audits is prerequisite for medical device manufacturing authorization.

Comprehensive post-market surveillance systems meeting MDR Articles 83–92 and EU AI Act Article 72 requirements including incident reporting and performance monitoring must be established before commercial launch. Early notified body engagement provides strategic guidance on conformity assessment requirements and reduces regulatory review risk.

11.4. User Training and Clinical Integration

User training programs should address initial requirements, competency assessment, and continuing education, specifically covering common error patterns in AI-generated segmentations, clinical scenarios requiring enhanced verification scrutiny, appropriate responses when segmentations are unreliable, and automation bias mitigation.

11.5. Future Expansion Opportunities

The technical foundation established through this project supports multiple pathways for expansion. The methodologies validated for radius and ulna segmentation transfer directly to other orthopedic structures including wrist carpal bones, ankle bones for distal tibia fractures, vertebrae for spinal surgery planning, and long bones for fracture fixation with anatomically appropriate adaptations.

Beyond anatomical segmentation, additional AI capabilities could support surgical planning including fracture pattern classification, fragment displacement quantification, reduction quality assessment, and surgical outcome prediction models.

■ References

■ References

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A. Appendix A: Training Configuration Details

A.1. Best Performing Model Configurations

ResU-Net v4 Configuration:

- Architecture: Classic-Res U-Net
- Initial features: 24
- Total parameters: 3,199,802 (3.2M)
- Optimizer: AdamW
- Learning rate: 0.0003 with CosineAnnealingLR scheduler
- Loss function: Combined CrossEntropy + Focal Loss (0.7:0.3 ratio)
- Regularization: Dropout 0.15, Weight decay 1×10^{-4}
- Training duration: 75 epochs, 25,308 seconds (7.03 hours)
- Batch size: 8
- Patch size: $128 \times 128 \times 128$ voxels
- Augmentation: None
- Hardware: NVIDIA RTX A6000 (48GB VRAM)

ResU-Net v5 Configuration:

- Architecture: Classic-Res U-Net
- Initial features: 20
- Total parameters: 2,223,312 (2.2M)
- Optimizer: AdamW
- Learning rate: 0.0005 with OneCycleLR scheduler
- Loss function: Combined CrossEntropy + Focal Loss
- Regularization: Dropout 0.15, Weight decay 2×10^{-4}
- Additional features: Multi-scale deep supervision, spatial attention
- Training duration: 80 epochs, 31,008 seconds (8.6 hours)
- Batch size: 8

nnU-Net Benchmark Configuration:

- Architecture: nnU-Net 3D Full Resolution (self-configured)
- Initial features: 32
- Total parameters: 31,000,000 (31M)
- Optimizer: SGD with momentum
- Learning rate: 0.1 with PolynomialLR decay
- Loss function: Combined Dice + CrossEntropy
- Training duration: 1000 epochs, 226,800 seconds (63 hours)
- Batch size: 2
- Post-processing: Largest component selection, hole filling, morphological closing

For comprehensive training details including failed experiments, hyperparameter exploration, debugging processes, and architectural design decisions, refer to [Appendix G](#) (Model Development Report). For complete experimental logs across all 150+ training runs, refer to [Appendix F](#) (Model Training Logs).

B. Appendix B: Performance Metrics Definitions

B.1. Dice Similarity Coefficient

The Dice Similarity Coefficient (DSC) measures overlap between predicted segmentation P and ground truth G as:

$$\text{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|} \quad (1)$$

Range: $[0, 1]$, where 1 is perfect overlap. Clinical threshold: $\text{DSC} \geq 0.85$ for acceptability.

B.2. Computational Metrics

Parameter Count: Total number of trainable parameters in the model, directly impacting memory requirements and computational complexity.

Training Time: Wall-clock time from training initialization to convergence, impacting development iteration speed.

Inference Time: Processing time per case including preprocessing and post-processing, impacting clinical workflow integration.

C. Appendix C: Dataset Statistics

C.1. Dataset Composition

Split	Cases	Percentage
Training	113	70%
Validation	24	15%
Test (Radius-only)	23	15%
Test (Multiclass)	19	15%
Total	162	100%

Table 3. Dataset Split Distribution

Note: The difference in test set sizes between radius-only (23 cases) and multiclass (19 cases) reflects dataset preparation iterations and case availability at different project phases.

D. Appendix D: Business Requirements Document

For comprehensive project planning including problem definition, proposed solution, success criteria, and acceptance testing framework, refer to the complete Business Requirements Document (BRD):

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/business_understanding/Final_BRD_Specifix.pdf

E. Appendix E: Data Preparation Report

For detailed data preparation methodology including the 8-step preprocessing pipeline, data quality assessment, patch extraction strategies, and HDF5 batching implementation, refer to the complete Data Preparation Report:

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/data_preparation/Data%20Preparation%20Report.pdf

F. Appendix F: Model Training Logs

For complete training logs documenting all 150+ experimental runs including configurations, performance metrics, and lessons learned, refer to the comprehensive model training log spreadsheet:

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/model_development/Model_log_final.xlsx

The spreadsheet contains two sections:

- **Radius Segmentation:** Experiments for radius-only (binary) segmentation task
- **Radius + Ulna Segmentation:** Experiments for multiclass segmentation task

G. Appendix G: Model Development Report

For comprehensive model development details including architectural variants, training configurations, hyperparameter exploration, per-case performance analysis, and failure mode identification, refer to the complete Model Development Report:

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/model_development/Model_Development_Report.pdf

H. Appendix H: Baseline Comparison Report

For detailed baseline architecture comparison including systematic evaluation of 3D U-Net, ResU-Net, V-Net, and nnU-Net with selection rationale and early experimental results, refer to the complete Baseline Comparison Report:

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/model_development/Baseline_Comparison_Expense_Report.md

I. Appendix I: Regulatory and Ethical Compliance

For comprehensive regulatory and ethical analysis including detailed compliance checklists, risk assessment matrices, and mitigation strategies, refer to the complete regulatory and ethical compliance documentation:

1. Comprehensive Report (Long Version):

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/ethical_and_legal_report/Legal%20and%20Ethical%20Report%20Specifix%203dBoneSegmentation.pdf

2. Executive Summary (Short Version):

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/ethical_and_legal_report/LEGAL%20AND%20ETHICAL%20FRAMEWORK%201to5page.pdf

3. Compliance Checklist (Excel):

https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-specifix-wrist-fracture-detection/blob/main/docs/ethical_and_legal_report/Legal_Ethical_Checklist.xlsx

Regulatory Classification Summary:

- **EU AI Act:** High-risk AI system (Article 6.1)
- **MDR Classification:** Class IIa medical device (Rule 11, Annex VIII)
- **Required Standards:** ISO 13485 (Quality Management), ISO 14971 (Risk Management)
- **Regulatory Pathway:** CE marking, clinical evaluation, notified body assessment
- **Human Oversight:** Mandatory surgeon review of all automated outputs
- **Post-Market Surveillance:** MDR Articles 83-92, EU AI Act Article 72

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